Practice Quiz: Planning — Designing Architectures in Yarn

Part A: Multiple Choice

1. Before beginning a crochet project, a crocheter must choose between flat vs. 3D, rows vs. rounds, and mesh vs. solid. In neural network terms, these choices are best described as: A) Training hyperparameters B) Architectural decisions that determine the topology and connectivity of the resulting structure C) Loss function selections D) Data preprocessing steps

2. The Euler characteristic of a sphere is 2, meaning that a crocheted amigurumi ball must: A) Have only increases and no decreases B) Have increases in the first half and matching decreases in the second half, so the total curvature budget equals 4 pi C) Be worked entirely in chain stitches D) Have the same stitch count in every round

3. Working in the round (as opposed to working in rows) creates a structure that is topologically equivalent to: A) A feedforward neural network — information flows in one direction only B) A recurrent neural network — each layer loops back and connects to its own beginning C) A transformer network with attention mechanisms D) A network with no connections at all

4. Front post double crochet stitches that reach around the post of a stitch two rows below are analogous to: A) Bias terms in a neural network B) Skip connections (residual connections) that allow information to bypass intermediate layers C) The learning rate of a training algorithm D) The activation function of a neuron

5. A crocheter making a gauge swatch before starting a project is performing: A) A pragmatic action — directly producing the desired fabric B) An epistemic action — reducing uncertainty about stitch dimensions before committing to the full project C) A random action with no planning value D) A communication action — sharing their model with others

6. A toroidal cowl (donut shape) has Euler characteristic 0. This means: A) No increases or decreases are ever needed B) The total positive curvature on the outer rim must be exactly canceled by negative curvature on the inner rim, so the overall curvature sums to zero C) The cowl must be flat D) Only chain stitches can be used
7. A granny square motif repeated across a blanket is structurally similar to: A) A dropout layer that randomly removes connections B) A recurrent feedback loop C) A convolutional filter — the same local pattern applied repeatedly across the surface D) A single fully connected layer with no repeated structure

Part B: Short Answer

1. You want to crochet a physical model of a feedforward neural network with an input layer of 6 neurons, a hidden layer of 3 neurons, and an output layer of 1 neuron. Describe your design: what is the foundation chain? How do you narrow from 6 to 3 to 1? What stitch operations represent the converging connections between layers? How could you use color or surface crochet to indicate different connection strengths?
2. Explain the planning difference between a flat blanket (Euler characteristic 1) and an amigurumi sphere (Euler characteristic 2) in terms of the "curvature budget." How does the Euler characteristic tell you, before you begin stitching, how much total increase and decrease work will be required? Give specific numbers for a single crochet project.
3. A crocheter is choosing between two projects: (A) a simple scarf in a stitch they know well, and (B) a sculptural hyperbolic coral piece requiring new techniques. Analyze this choice using the Active Inference concept of expected free energy, identifying the pragmatic value and epistemic value of each option. Under what circumstances would each project be the better choice?